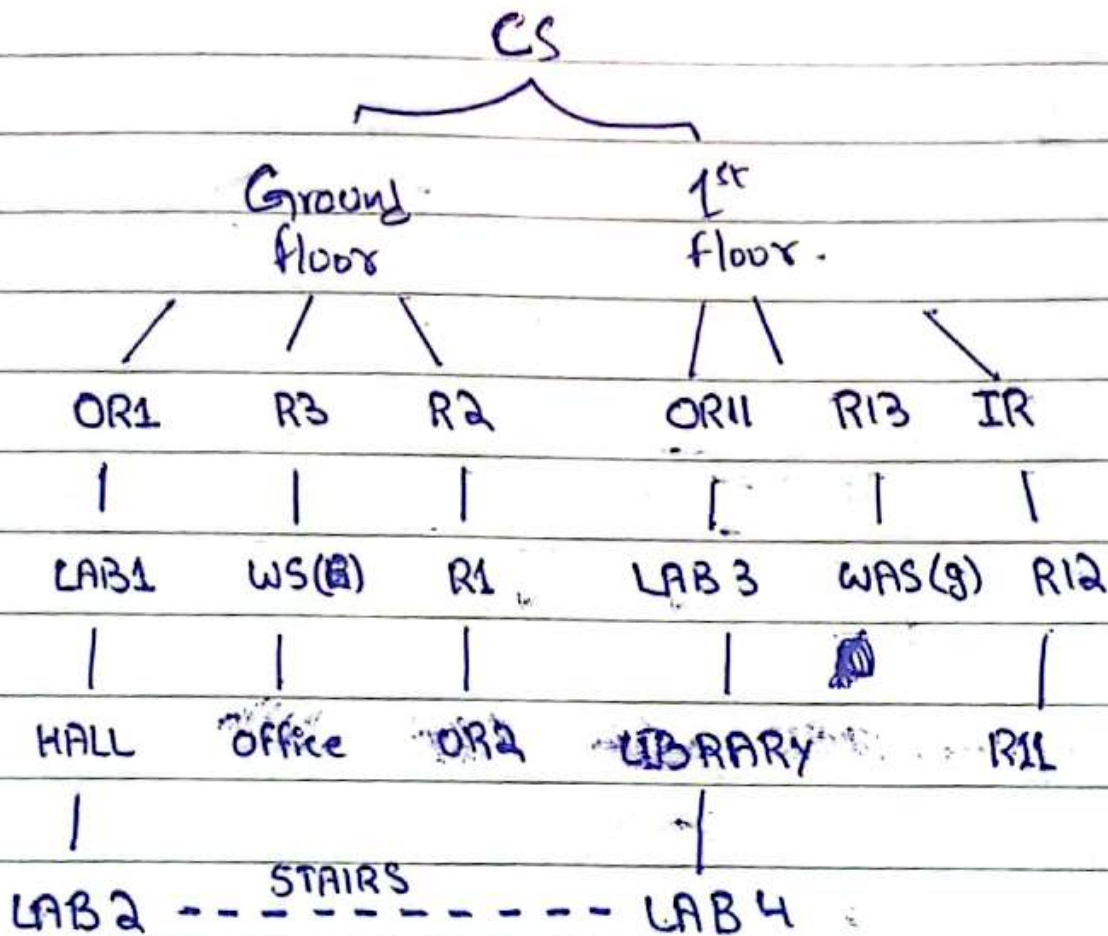


GRAPH (E) CS DEPARTMENT.



TRANSITION TABLE :-

CS	Ground floor, 1 st floor
Ground floor	OR1, R3, R2, CS
First floor	OR11, R13, IR, CS
OR1	LAB 1, ground floor
R3	WS(B), ground floor

Date: _____

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 M T W T F S S


R2

Lab 1

WS(b)

ORA

Hall

office

ORA

Lab 2

ORII

R13

IR

Lab 3

Was (g)

R12

Library

R11

Lab 4

STAIRS

R1, ground floor

Hall, OR1

office, R3

ORA, R2

Lab 1, Lab 2

WS(b)

R1

Hall, stairs

First floor, Lab 3

First floor, WS(g)

First floor, R12

ORII, Library

R13, Library

R11, IR

Lab 3, Lab 4

R12

Library, stairs

LAB 2, LAB 4

PROBLEM FORMULATION:-

Initial state = Lab 2

Goal state = Lab 4

State space = All possible locations in CS department

Transition model = possible movements between states

Path cost = distance

BFS ~~structure~~ solution:-

Initialization:-

Q = [LAB 2]

V = [LAB 2]

Deq. LAB 2 & expand:-

Q = [HALL, STAIRS]

V = [LAB 2, HALL, STAIRS]

Deq. HALL & expand:-

Date:

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M T W T F S S

~~Q~~ Q = [STAIRS, LAB 1] ~~LAB 2~~

V = [LAB 2, LAB 1, STAIRS, HALL]

Deq. STAIRS & EXPAND:

Q = [LAB 1, LAB 4]

V = [LAB 2, LAB 1, STAIRS, HALL, LAB 4]

Deq. LAB 1 & EXPAND:

Q = [LAB 4, ORI]

V = [LAB 2, HALL, STAIRS, LAB 1, LAB 4, ORI]

Deq. LAB 4 & EXPAND: (Goal state found)

Q = [ORI, LIBRARY, ~~STAIRS~~]

V = [LAB 2, HALL, STAIRS, LAB 1, LAB 4, ORI, LIBRARY]



Date: _____

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M T W T F

SHORTEST PATH :-

LAB 2



Steps



LAB 4